Retaliatory Evictions: Distributed-Lag Evidence

Philly Evictions Project

2026-03-04

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1 Overview

This writeup now reports only the retaliatory timing analysis from `r/retaliatory-evictions.r`:

- distributed-lag event-time regressions (`filed_eviction` on lags/leads of complaint indicators)
- bandwidth diagnostics around eviction filings
- distributed-lag heterogeneity by building-level `% Black` split

Complaint-suppression analysis was intentionally removed from this workflow and now lives in `r/comp-statics-annual-race.R` and `writeups/comp-statics-annual-race.qmd`. Unless explicitly overridden in the script call, outputs shown here are pre-COVID only (`<= 2019`).

2 Empirical Bayes Filing-Rate Memo

The maintenance regressions below now use `filing_rate_eb_pre_covid` as the default filing-intensity measure. This is the building-level Empirical Bayes (Gamma-Poisson) posterior mean pooled over pre-2019 years:

$$y_{it} \mid \lambda_i \sim \text{Poisson}(E_{it}\lambda_i), \qquad \lambda_i \sim \text{Gamma}(\alpha, \beta),$$

with sufficient statistics

$$Y_i = \sum_t y_{it}, \qquad E_i = \sum_t E_{it},$$

and posterior mean

$$\mathbb{E}[\lambda_i \mid Y_i, E_i] = \frac{\alpha + Y_i}{\beta + E_i}.$$

In the production pipeline (`r/make-analytic-sample.R`, section 6b), the city prior is estimated by method-of-moments; ZIP-specific priors are used when supported, otherwise the city prior is used as fallback. The raw long-run filing rate remains available for robustness and is reported in the appendix of this writeup.

Reference memo for reuse: `docs/notes/eb-filing-rate-memo-2026-03-04.md`.

3 Estimating Equations

3.1 Distributed lag (main)

For complaint stem $c \in \{\text{complaint}, \text{severe}, \text{non_severe}\}$:

$$Y_{it} = \sum_{k=1}^{h} \beta_{k,c}^{-} D_{i,t-k}^{c} + \beta_c^0 D_{i,t}^{c} + \sum_{k=1}^{h} \beta_{k,c}^{+} D_{i,t+k}^{c} + \alpha_i + \gamma_t + \varepsilon_{it},$$

where:

- $Y_{it} = \mathbb{1}[\text{filed_eviction}_{it} = 1]$
- $D_{it}^{c} = \mathbb{1}[\text{filed_}c_{it} = 1]$
- α_i: PID fixed effects, γ_t: period fixed effects
- SEs clustered at PID

3.2 Race heterogeneity split

Define:

$$G_{it} = \mathbb{1}[\text{infousa_pct_black}_{it} \geq c_{\text{pre}}],$$

where c_{pre} is computed from pre-2020 $(PID, year)$-unique observations using the configured rule (default: pre-COVID mean).

The same distributed-lag equation is estimated on high- and low-G_{it} subsamples.

4 Data and Sample

- Unit: PID × quarter
- Default window: 2011–2019
- Inputs: rental building panel, analytic PID filter, eviction filings + address crosswalk, and `bldg_panel_blp` tenant composition
- Eviction filing sample restrictions (implemented in `r/retaliatory-evictions.r`):
  - keep only filings with unique parcel crosswalk matches (`num_parcels_matched == 1`)
  - drop commercial filings (`commercial != TRUE`)
  - drop non-residential filings (`non_residential != TRUE`)
  - keep only non-payment-of-rent filings (`a == TRUE` in `evictions_clean`)
  - apply the configured year window (default pre-COVID: 2011–2019)
- `Full` complaint indicator: any complaint filing in the period.
- `Severe` complaint indicator: any period complaint in the severe bundle (`heat`, `fire`, or `property_maintenance`), consistent with `total_severe_complaints` in the panel.
- `Non-Severe` complaint indicator: any complaint in the non-severe bundle used in the script.

This aligns the filing sample with the non-payment focus in Lodermeier (2025) and with the eviction data README note that complaint-derived fields are consistently available from 2006 onward, with a complaint format change in 2011.

Table 1: Eviction filing sample restrictions used in this analysis

Step	Rows	Share of raw
raw_evictions_clean_rows	796,440	100.0%
unique_parcel_xwalk_match	535,757	67.3%
drop_commercial_cases	534,826	67.2%
drop_non_residential_cases	530,654	66.6%
keep_nonpayment_of_rent_cases_a_flag	336,502	42.3%
keep_year_between_2011_and_2019	174,940	22.0%

Table 2: Eviction filing decomposition within quarters windows

Bandwidth	Same period	Within bandwidth	Within bandwidth, not same period	Non-retaliatory
1	14.0%	26.3%	12.3%	73.7%
2	14.0%	32.8%	18.8%	67.2%

5 Bandwidth Diagnostics

6 Distributed-Lag Results

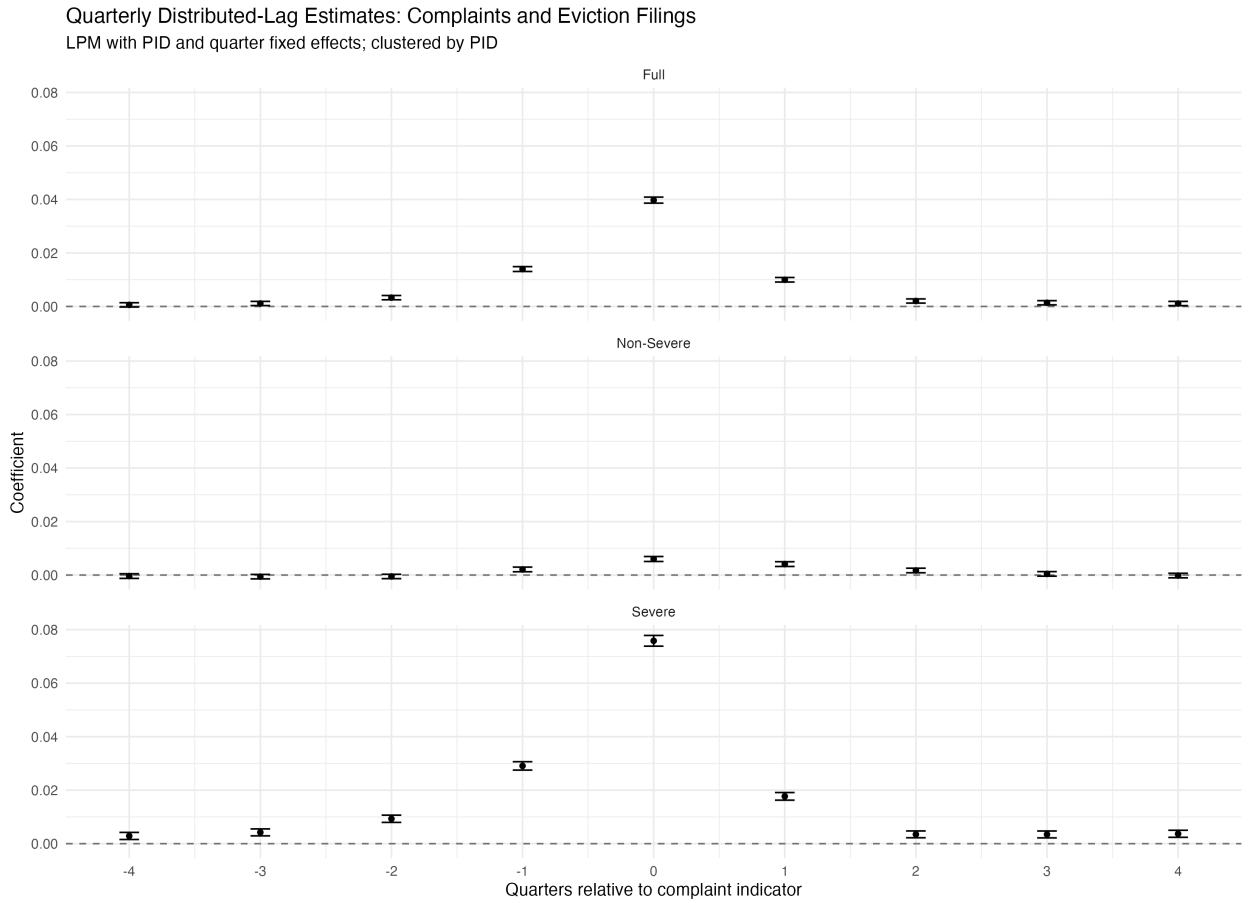

Figure 1: Distributed-lag coefficients (pre-COVID): effect of complaint indicators at lag/lead k on eviction filing indicator. Panels are labeled Full, Non-Severe, and Severe. PID and period FE; SE clustered by PID.

7 % Black Heterogeneity in Distributed Lags

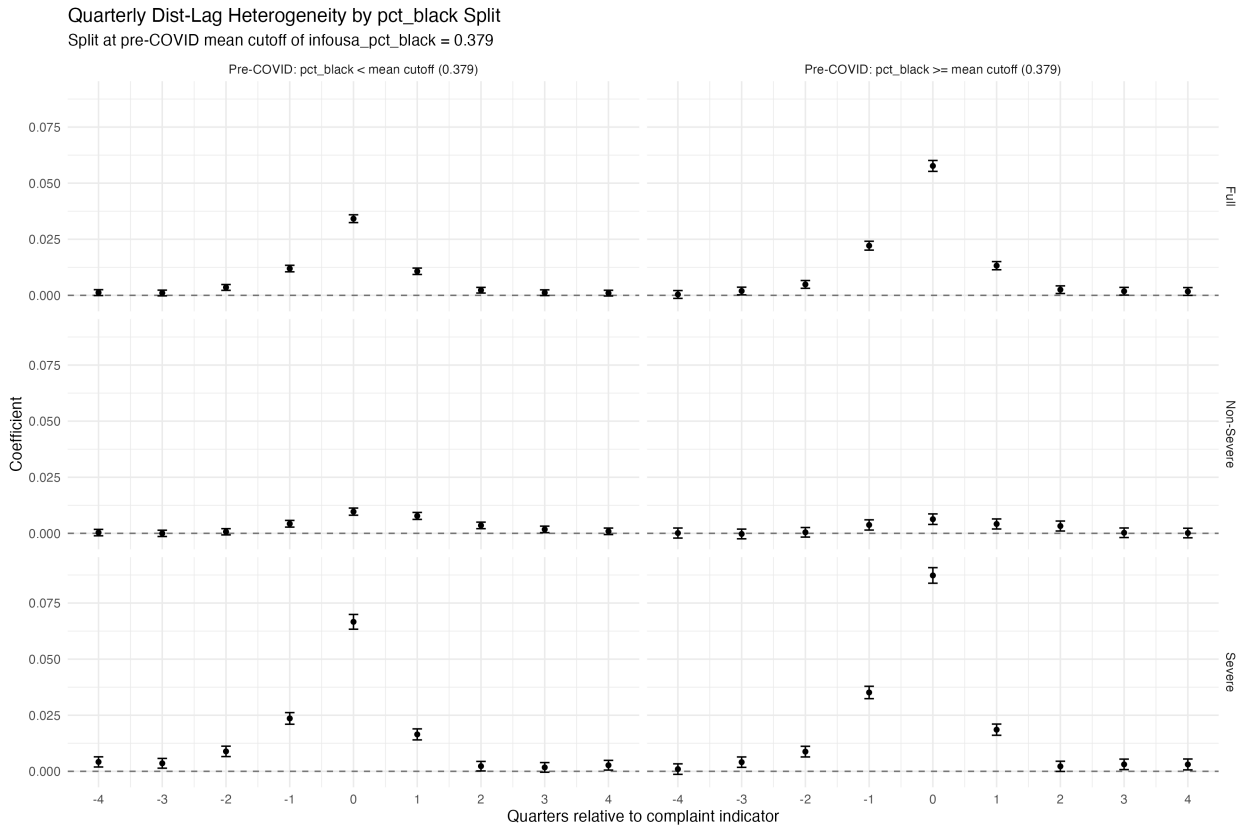

Figure 2: Distributed-lag coefficients by high/low pct black split (cutoff from pre-COVID distribution rule configured in script).

8 One-Month Back-Rent Regressions

Case assignment for this block:

- Outcome: `one_month_back_rent = 1` when `total_rent / ongoing_rent` is within `1 ± tol` (default `tol = 0.10`).
- `retaliatory_severe = 1` if an eviction case's PID-period has a severe complaint in the same period or within `± bandwidth` periods (default bandwidth is `2` quarters).
- `high_black_split = 1` if building `infousa_pct_black` is above the pre-COVID mean cutoff.
- Defendant race category from case-level imputation (`r/impute-race.R` output): `black` if `case_p_black >= 0.8`, `white` if `case_p_white >= 0.8`, and `unknown` otherwise.
- In the defendant-race interaction model, `white` is the reference category.

Table 3: Pre-COVID case assignment by severe-complaint proximity

Status	Cases	Share
non_retaliatory	86770	61.7%
same_period_severe	25554	18.2%
within_bw_severe_not_same_period	28341	20.1%

Table 4: Regression sample summary (pre-COVID; PID + year FE specs)

Spec	N	Mean(1 month rent)	Same-period severe share	Within-bw severe (not same period) share	Above-mean pct black share	Defendant race: Black	Defendant race: White	Defendant race: Unknown	Defendant race impute-ok share
Status dummies only	115107	0.1898	18.2%	20.1%	55.6%	–	–	–	99.5%
Status x building above-mean pct black	90737	0.1885	18.4%	21.3%	55.6%	–	–	–	99.5%

Dependent Variable:	one_month_back_rent	
Model:	(1)	(2)
Variables		
Same-period severe	0.017***	-0.011
	(0.006)	(0.012)
One-quarter pre/post severe	-5.95×10^{-5}	-0.019*
	(0.006)	(0.010)
Above-mean building pct black		-0.0003
		(0.010)
Same-period severe x above-mean building pct black		0.046***
		(0.016)
One-quarter pre/post severe x above-mean building pct black		0.032**
		(0.013)
% cases filed in quarter of complaint	18.2	18.4
% cases in one-quarter pre/post window	20.1	21.3
Fixed-effects		
PID	Yes	Yes
year	Yes	Yes
Fit statistics		
Observations	115,107	90,737
R^2	0.29134	0.29007
Within R^2	0.00025	0.00063

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

8.1 Poisson (PPML) Regressions: Months Back Rent

The same three specifications are repeated using Poisson (PPML) with `months_back_rent` (= `total_rent / ongoing_rent`, continuous) as the outcome. Coefficients are log-multiplicative; SEs clustered by PID.

Dependent Variable:	months_back_rent	
Model:	(1)	(2)
Variables		
Same-period severe	-0.027**	0.025
	(0.013)	(0.026)
One-quarter pre/post severe	0.005	0.028
	(0.010)	(0.017)
Above-mean building pct black		0.017
		(0.023)
Same-period severe x above-mean building pct black		-0.080**
		(0.031)
One-quarter pre/post severe x above-mean building pct black		-0.047**
		(0.022)
% cases filed in quarter of complaint	18.2	18.4
% cases in one-quarter pre/post window	20.1	21.3
Fixed-effects		
PID	Yes	Yes
year	Yes	Yes
Fit statistics		
Observations	115,043	90,679
Squared Correlation	0.48646	0.49284
Pseudo R^2	0.16481	0.16303
BIC	605,399.7	465,151.8

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

8.2 OLS Log-Rent Regression: log(Total Rent) | log(Ongoing Rent)

Outcome is `log(total_rent)` with `log(ongoing_rent)` entered as a continuous covariate. Conditioning on log ongoing rent removes the mechanical correlation between total back rent and rent level across tenants. The coefficient on `retaliatory_status` therefore captures the percentage difference in total back rent *claimed* for retaliatory versus non-retaliatory cases, holding the monthly rent fixed. A negative coefficient means landlords file for less back rent in retaliatory cases — i.e., the filing is not primarily about recovering arrears.

Dependent Variable:	log(total_rent)	
Model:	(1)	(2)
Variables		
Same-period severe	-0.025**	0.018
	(0.012)	(0.024)
One-quarter pre/post severe	0.005	0.029*
	(0.008)	(0.015)
log(ongoing_rent)	0.693***	0.695***
	(0.019)	(0.021)
Above-mean building pct black		-0.001
		(0.017)
Same-period severe x above-mean building pct black		-0.068**
		(0.028)
One-quarter pre/post severe x above-mean building pct black		-0.051***
		(0.019)
% cases filed in quarter of complaint	18.2	18.4
% cases in one-quarter pre/post window	20.1	21.3
Fixed-effects		
PID	Yes	Yes
year	Yes	Yes
Fit statistics		
Observations	112,729	88,859
R^2	0.49715	0.50065
Within R^2	0.04103	0.04208

Clustered (PID) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

9 Equilibrium Maintenance Analysis

This section examines whether landlords adjust maintenance investment (proxied by permit activity) in response to eviction pressure and tenant composition, and whether permit activity responds dynamically to severe complaints.

9.1 Baseline Maintenance Permit Regressions

Poisson (PPML) regressions of `total_permits` on building-level eviction filing intensity and racial composition, with `log(total_units)` as offset. Main specifications use EB pre-COVID filing intensity (`filing_rate_eb_pre_covid`) with:

- a high-filing dummy (`>= 15%`)
- filing bins: `[0,1)`, `[1,5)`, `[5,10)`, `[10,20)`, `[20,inf)` (all on EB rates)

The rent-augmented variants add mean inflation-adjusted log rent (within PID over 2011–2019) to partial out rent-driven maintenance incentives.

Dependent Variable:	total_permits			
Model:	(1)	(2)	(3)	(4)
Variables				
above_mean_pct_black_btTRUE	-0.375***	-0.327***	-0.379***	-0.335***
	(0.007)	(0.009)	(0.007)	(0.009)
high_filing_eb	-0.153***	-0.101***		
	(0.012)	(0.012)		
mean_log_med_rent_inf_adj_PID		0.376***		0.383***
		(0.010)		(0.010)
filing_bin_eb = [1,5)")			0.093***	-0.003
			(0.008)	(0.010)
filing_bin_eb = [5,10)")			-0.154***	-0.130***
			(0.009)	(0.011)
filing_bin_eb = [10,20)")			-0.130***	-0.090***
			(0.011)	(0.012)
filing_bin_eb = [20,inf)")			-0.219***	-0.179***
			(0.015)	(0.016)
Fixed-effects				
building_type	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
quality_grade	Yes	Yes	Yes	Yes
year_blt_decade	Yes	Yes	Yes	Yes
num_units_bin	Yes	Yes	Yes	Yes
GEOID	Yes	Yes	Yes	Yes
has_rent_data	Yes		Yes	
Fit statistics				
Observations	1,703,026	750,270	1,683,789	743,824
Squared Correlation	0.31379	0.25057	0.32030	0.25933
Pseudo R^2	0.20782	0.26005	0.20954	0.26192
BIC	1,144,296.2	615,091.5	1,121,025.3	602,300.0

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

9.2 Landlord Response to Severe Complaints

The following coefplot shows estimates from a PPML regression of `total_permits` on contemporaneous, lagged (past quarters), and lead (future quarters) severe complaint counts, with PID and year fixed effects. Lead terms serve as pre-trend checks.

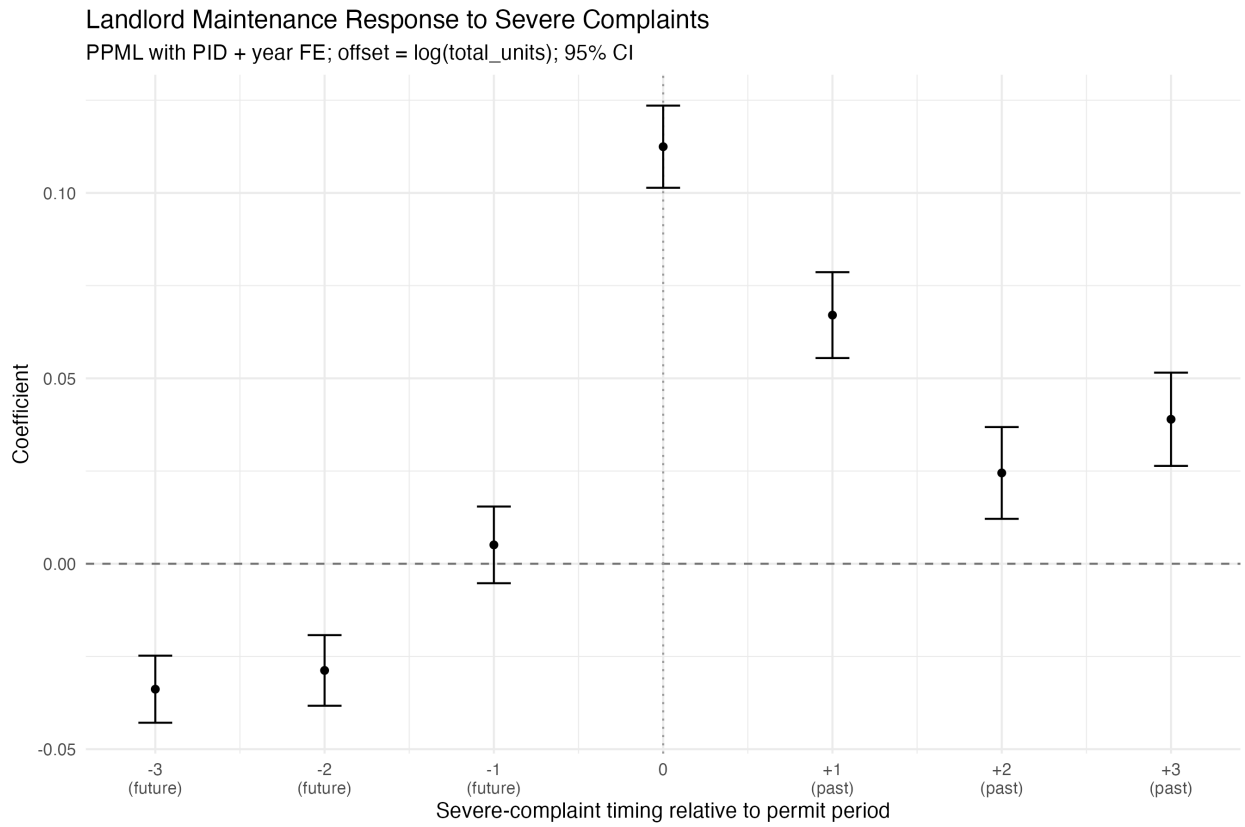

Figure 3: Landlord maintenance response to severe complaints: PPML event-study. Negative timing = complaints occur in future periods relative to permit outcome (pre-trend check); positive timing = past complaints. PID + year FE; offset = log(total_units); 95% CI.

10 Notes

- Raw long-run filing-rate maintenance regressions are moved to the appendix section below.
- This writeup intentionally excludes same-period, targeting, LP-DiD, and complaint-suppression blocks.
- For complaint-suppression specifications and interpretation, use `writeups/comp-statics-annual-race.qmd`

11 Appendix: Raw Filing Robustness

These regressions mirror the baseline maintenance models but replace EB filing intensity with raw long-run pre-2019 filing rate (same 15% high-filing cutoff and same bins), for robustness only.

Dependent Variable:	total_permits			
Model:	(1)	(2)	(3)	(4)
Variables				
above_mean_pct_black_btTRUE	-0.373***	-0.326***	-0.362***	-0.316***
	(0.007)	(0.009)	(0.007)	(0.009)
high_filing_raw	-0.211***	-0.137***		
	(0.013)	(0.014)		
mean_log_med_rent_inf_adj_PID		0.375***		0.352***
		(0.010)		(0.010)
filing_bin_raw = [1,5)")			-0.150***	-0.184***
			(0.010)	(0.011)
filing_bin_raw = [5,10)")			-0.197***	-0.137***
			(0.009)	(0.011)
filing_bin_raw = [10,20)")			-0.217***	-0.151***
			(0.013)	(0.014)
filing_bin_raw = [20,inf)")			-0.324***	-0.232***
			(0.015)	(0.016)
Fixed-effects				
building_type	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
quality_grade	Yes	Yes	Yes	Yes
year_blt_decade	Yes	Yes	Yes	Yes
num_units_bin	Yes	Yes	Yes	Yes
GEOID	Yes	Yes	Yes	Yes
has_rent_data	Yes		Yes	
Fit statistics				
Observations	1,703,026	750,270	1,703,026	750,270
Squared Correlation	0.31361	0.25042	0.31484	0.25360
Pseudo R^2	0.20788	0.26008	0.20841	0.26058
BIC	1,144,210.6	615,062.0	1,143,500.9	614,705.9

IID standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*